

CONTEXTGRAPH: MEASURING TRANSFER FROM EXPERIENCE TO VERIFIED CODE REPAIRS

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ABSTRACT

Long-term memory can change what a coding agent reads without improving the patch it submits. We study this gap using ContextGraph, a system that turns prior repair evidence into rules and retrieves them through dense, lexical, and graph channels. Our evaluation separates memory provenance, actual delivery, agent execution, and official repair verification. Existing official reports for two completed SWE Context Bench Related-Lite conditions record 18 resolved predictions each on the same 99 tasks, with six wins unique to each condition. The recorded paired difference is 0 percentage points (95% bootstrap interval $[-7.07, 7.07]$). A completed contemporaneous Control resolves 17/99; Flat and Card each gain only one observed repair over Control, with exact paired $p = 1$ for both comparisons. An extended audit identified one appended historical Docker log; an independent frozen-verifier recheck of the identical prediction reproduces the original failure, with no outcome changed. Actual memory is delivered on all tasks, while the correct procedure card reaches 98 of 99 targets. An adaptive card revision resolves 16/99 with correct cards delivered on 97/99, without a significant paired gain. These observations do not establish a general limit on agent memory. We also prepare a separate SWE-bench Verified experiment that builds memory from 350 training tasks and compares no memory, flat rule retrieval, and graph rule retrieval on a fixed 150-task test set. This draft reports the audits, the retained provenance limitation, and the new experiment’s design; it does not yet report a test-set gain from the latter.

Draft status, 7 September 2026. This is a research draft, not a completed submission. Related-Lite first predictions have checked frozen-verifier evidence, including one independent recheck for the historical log exception described below. The separate Verified 350/150 inputs are frozen; execution stopped on an agent-framework error, and no formal effect estimate is available. Historical results from other repository experiments are not pooled into either denominator.

1 INTRODUCTION

Coding agents must locate relevant code, construct an explanation of a failure, edit a repository, and verify a repair. Prior experience could help at each stage, but a retrieval hit is not itself a successful repair. A memory system may return a correct example for the wrong software version, fail to deliver its output to the agent, or consume enough interaction budget to displace useful testing. Conversely, a low repair rate does not establish that the retrieved knowledge was unusable: the surrounding tool or evaluation protocol may have failed.

We investigate this distinction in ContextGraph, a long-term memory layer for coding agents. Its broader architecture combines typed experience nodes with lexical, embedding, and Personalized PageRank retrieval. The present study uses two distinct settings. The first is a retrospective audit of a completed repository-card intervention on SWE Context Bench Related-Lite (Zhu et al., 2026). The second is a new, deliberately narrower intervention on SWE-bench Verified (Jimenez et al., 2024): fixed rules derived from training issues and their official patches, with graph retrieval isolated from the rest of the retrieval stack. The latter is supervised memory construction with a frozen solver, not reinforcement learning or evidence of self-improvement from the solver’s own successful attempts.

Our current contributions are an evidence-linked evaluation procedure, an audited result for the repository-card intervention, and a controlled experiment specification for graph-based transfer. The scientific question is which forms of prior repair evidence improve verified outcomes after their delivery, cost, and source eligibility have been established.

2 RELATED WORK

Experience and graph retrieval. ExpeL extracts reusable natural-language insights from agent experience without requiring updates to proprietary model weights (Zhao et al., 2024). HippoRAG combines knowledge graphs and Personalized PageRank for associative retrieval (Gutiérrez et al., 2024). These works motivate ContextGraph’s representation and retrieval choices; they do not establish that the same mechanisms improve a coding agent’s repair success under a fixed interaction budget. MOOSEDev models typed project records and supersession, evaluating structured questions over project documentation with an LLM judge (Adam, 2026); that endpoint differs from executable repair verification. SWE-agent demonstrates the importance of the agent–computer interface for software engineering (Yang et al., 2024), making interface fidelity part of our experimental specification.

Skills, working memory, and transfer. CODESKILL trains a skill-management mechanism while retaining a fixed coding policy; its reported Verified setup uses 350 training tasks and 150 evaluation tasks (Li et al., 2026). Our split size follows the same general design, but we do not claim to reproduce its task IDs, optimization procedure, or results. SWE-MeM studies adaptive working-memory management for long-horizon coding (Gao et al., 2026), whereas our new experiment supplies a fixed cross-task packet before solving. Agent Memory Distillation studies hierarchical teacher memory for smaller agents (Kim et al., 2026b); our experiment uses neither a small-model distillation objective nor teacher-weight training. Memory Transfer Learning distinguishes memory representations and reports that abstract knowledge can transfer more effectively than concrete traces in its evaluated settings (Kim et al., 2026a). This motivates a rule-based transfer hypothesis, without guaranteeing its outcome in our repositories.

Process measurements are intermediate outcomes. The Working Set studies coherence in repository-scale agent tasks (Mohammadi et al., 2026). Measure Before You Manage separates working-memory management and its evaluation (Chen et al., 2026a); its small policy-evaluation populations and incomplete official repair verification limit direct comparison with a fully graded SWE campaign. ClawTrace relates skill distillation to trajectory costs (Yuan et al., 2026). We use these works to motivate measurements of delivery, repeated errors, and consumption cost, while keeping official repair verdicts as the outcome of interest. We do not turn their setting-specific findings into universal claims about which memories help or harm.

Saha’s cue-anchored memory study evaluates deterministic delivery on one coding task and repeated-compaction fact survival (Saha, 2026). Delivery and its audit are therefore established concerns. Our retention analysis reconstructs the original responses selected by a fixed history processor across 198 first trajectories, linked to their original predictions and frozen-verifier evidence; it neither introduces reinjection nor estimates its causal benefit.

Recent scaffold optimization also uses persistent beliefs: BCO calibrates a document against observed outcomes (Chen et al., 2026b). Its two memory-QA rows select the best held-out result among three candidates, a disclosed selection step that limits interpreting them as untouched test estimates. CANOPY instead trains the policy through outcome-only reinforcement learning (Pu et al., 2026); it is a distinct intervention from our frozen-model memory injection.

3 EVALUATION MODEL

For task i and condition a , let M_{ia} denote the supplied memory, D_{ia} actual delivery, T_{ia} the trajectory, and Y_{ia} the official verifier’s binary result. These objects are linked rather than inferred from one another:

$$\text{eligible sources} \longrightarrow M_{ia} \longrightarrow D_{ia} \longrightarrow T_{ia} \longrightarrow \text{submitted patch} \longrightarrow Y_{ia}.$$

Each link has a separate check. Source IDs must satisfy the frozen split; delivered content must appear in the actual model input or matching tool response; submitted patches and trajectories must match their recorded hashes; and a positive result must agree with both the task report and the harness summary. An unbound or protocol-invalid attempt lacks an admissible outcome. Explicit empty-submission and verifier-setup-failure labels follow the matching frozen policy and are distinguished from tested failures.

Fixed universes. An analysis uses its preregistered task list, never the intersection of whichever receipts happen to exist. If a campaign is incomplete, we report completion and missingness against that universe; we do not present a complete paired effect estimate. An infrastructure-only verifier repair is recorded separately and bound to the unchanged prediction. A valid failed attempt is not replaced by a later successful solver run.

Delivery gates. Before interpreting a memory condition, at least 95% of targets must receive the expected memory and there must be no phase-based refusal. For the Related-Lite card condition, the expected card must match the target repository. Connection errors and an unbound string in a log do not count as delivery. Passing this gate establishes exposure, not relevance, correctness, or causal mediation through the memory.

Cost and protocol. Solver call counts, submitted completion status, and repair verdicts are different measurements. We record actual provider input/output tokens and charges where available. In local SWE-agent 1.1.0 diagnostics, retokenizing only assistant text omitted tool-call output from the reported output-token total, and local estimated costs differed from provider receipts. We therefore do not interpret unaudited legacy cost totals as measured savings. The local runtime also preserves the provider’s returned reasoning fields across tool turns and checks the options actually sent on the wire. These are protocol controls; the present observations do not establish a causal repair-rate effect of any individual protocol change.

4 COMPLETED RELATED-LITE AUDIT

Setting. The completed conditions use the same frozen 99-task Related-Lite manifest and the same frozen SWE Context Bench verifier. The flat condition retrieves reference patches from the existing source pool. The original card condition (Card-v1, denoted Card) adds a repository-specific procedure card: test commands, environment repair conditions, reproduction routines, and workflow information. Its sources include pool-derived evidence and environment recipes checked in benchmark containers. This is an intervention on repository environment knowledge, not an isolated test of graph traversal. The actual retrieved patch sets can vary with agent-generated queries, so identical serving logic does not imply byte-identical patch inputs across runs.

Saved solver settings. All 594 first trajectories declare SWE-agent 1.1.0 at one Git commit and SWE-ReX 1.4.0. Selected settings agree across the six conditions: model alias `openai/deepseek-v4-pro`, temperature 0, top- p 1, configured input/output limits of 100,000/8,192 tokens, 500-call/\$30 per-task limits, and LastNObservations with $n = 5$. No fallback is configured. The version recorder reads Git HEAD without checking a dirty tree; these declarations do not attest historical solver file contents, the complete interpreter/dependency state, or every wire request.

Receipt audit. All 99 receipts in each condition pass task-identity, literal-boolean verdict, trajectory/prediction hash, and report/summary consistency checks. For nonempty patches, the initial audit also checks the frozen harness evidence inventory, its file hashes, recorded offline Docker attestations, and runtime identity before and after verification. The harness commit is recorded in the accompanying evidence index. One task has a later verifier receipt, but its original receipt is already a valid failure; the analysis retains that original result.

An extended check of raw Docker logs flags Flat’s `matplotlib-23687`: its original 15 log records reproduce the receipt’s exact hash, but 21 later execution records have been appended. The original official report is unchanged and records failure (3/4 fail-to-pass and 791/791 pass-to-pass tests). The relationship of the appended execution records to that receipt is not fully established. A separate

Table 1: Audited Related-Lite first-prediction outcomes. The historical 88-task source-overlap sensitivity set does not replace the full 99-task denominator. Differences and exact two-sided McNemar tests are versus the contemporaneous Control on all 99 tasks. The historical Flat log exception remains disclosed in the text.

Condition	Fixed 99	Sensitivity 88	Difference	Exact p
Control	17/99	16/88	Reference	—
Flat	18/99	18/88	+1.01 pp	1.000
Card	18/99	15/88	+1.01 pp	1.000
Graph	12/99	12/88	−5.05 pp	0.227
Rules	13/99	12/88	−4.04 pp	0.424
File list	9/99	8/88	−8.08 pp	0.057

diagnostic verification of the identical prediction reproduces the original failure and passes the raw-log, frozen-runtime, report, and input-hash checks. It neither reruns the solver nor replaces the original outcome. We attach this independent evidence to that first prediction and retain the original provenance exception: the recheck does not reconstruct the historical execution. All reported first-prediction outcomes remain unchanged.

Verdict semantics. This frozen Related-Lite harness requires all fail-to-pass tests to pass for resolved, irrespective of pass-to-pass failures. Across the six original conditions, replaying all 287 complete recorded before/after comparisons exactly reproduces the reports; 15 setup failures and 292 empty predictions remain separate. Four officially resolved predictions contain a recorded pass-to-pass transition from PASSED to FAILED. Pass-to-pass success also admits some unchanged skipped, failed, or missing statuses. Thus neither label guarantees all regression tests passed. The logs contain parsed status maps, not complete raw test stdout; official scores remain unchanged, and these semantics are not transferred to Verified.

Outcomes. On the fixed universe, both conditions resolve 18/99 tasks (Table 1). Twelve successes are shared, six belong only to Flat, and six only to Card. The two-sided exact McNemar test gives $p = 1$, and the paired 95% bootstrap interval for Card minus Flat is $[-7.07, 7.07]$ percentage points. The interval is compatible with effects in both directions; lack of significance is not evidence of equivalence. These tasks have been used in multiple prior interventions, so this is an exploratory audit rather than a newly untouched confirmatory evaluation.

Source-overlap sensitivity. The complete historical 11-task near-duplicate exclusion list defines an 88-task sensitivity set. Flat resolves 18/88 and Card 15/88; the discordant counts are six versus three, with $p = 0.508$ and interval $[-10.23, 3.41]$ percentage points. Three Card-only successes on the full set are flagged by the prior patch-overlap scan with Jaccard similarity 1. A subsequent screen of all 693 original trajectories finds same-repository added-line Jaccard ≥ 0.5 on 17/99 targets in Flat, Card, Graph, File list and Card-v2, and 14/99 in Rules; seven detected targets lie outside the historical list (Appendix A). These overlap observations do not prove cheating or redefine the sensitivity set, and neither audit certifies all contamination channels.

Actual exposure and mechanism checks. The tool-response census finds patch memory on 99/99 targets in each condition, no missing trajectories, and no phase refusals. The correct frozen repository card appears exactly in 98/99 Card trajectories. The exception is a scikit-learn task that receives a Matplotlib card; we retain its result in the fixed universe. The registered 95% exposure threshold is met, while this single routing failure remains part of the intervention’s observed behavior. All 198 debug logs are present, with no matches to the enumerated rate-limit error patterns; this narrow log check does not rule out every provider-side failure.

A post-observation reconstruction of all 198 first trajectories distinguishes delivery from retention. Using the pinned history processor and each saved configuration (`last_n_observations`, $n = 5$), the original served responses remain in a median of five recorded assistant-action prefixes per task in both arms, compared with median trajectory lengths of 497 for Flat and 495 for Card. Neither arm retains an original response at its last recorded action prefix. All 1,426 direct processor boundary

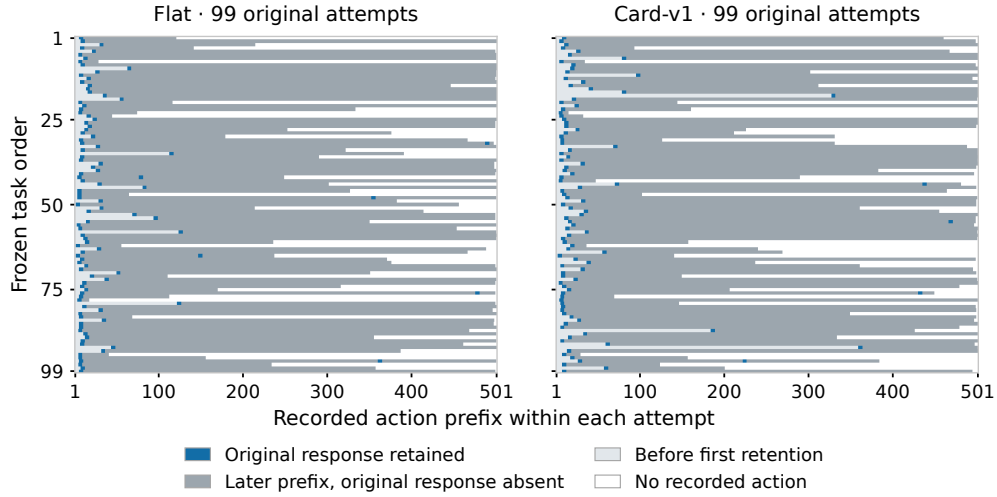


Figure 1: Original memory-response retention across all 198 first trajectories. Rows use the same fixed task-ID order in both panels. Blue marks a recorded input prefix retaining any original response; gray distinguishes before-first-retention and later absence. White is outside the recorded sequence. No outcomes determine row order or inclusion; absence here does not establish forgetting.

checks agree with the reconstruction. This measures original response-content retention, excluding paraphrases or copies elsewhere; it does not reconstruct every wire request or establish forgetting or a causal effect. Figure 1 shows every recorded sequence. First retention ranges from prefixes 3–125 for Flat and 3–359 for Card, so five retained prefixes does not mean the first five actions.

The hard `exit_cost` label occurs on 32 Flat tasks and 41 Card tasks, so Card does not meet its registered < 32 threshold. Both conditions have a median saved model-call counter of 501 under the legacy post-response call-limit behavior. The label can reflect the call cap, and must not be read as 41 dollar-budget overruns. An audit of all 198 first trajectories reproduces the archived test-output proxy: 81 Flat tasks and 73 Card tasks match, with conditional median first-match indices of 128 and 98 assistant messages. The proxy accepts failed tests, zero-test runs, and standalone OK prints; it is not a first-passing-test measure. Card meets the numerical timing threshold under this proxy but fails its coverage threshold. A conservative direct-runner and positive-execution-summary census recognizes 78/99 Flat tasks and 67/99 Card tasks, with conditional median indices of 116.5 and 97 recorded trajectory entries. Among the 60 tasks recognized in both conditions, Card is earlier on 38 and later on 22. This census omits some real tests in unsupported command forms; its step counter also differs from the archived counter, and counted unittest loader placeholders require the sensitivity check below. Conditional timing and incomplete recognition do not establish an overall speedup, test relevance, or a cause of the unchanged repair score.

The available gold-file call recognizer matches 61/99 Flat tasks and 49/99 Card tasks, with conditional median indices 212 and 165 assistant messages. Card meets this numerical median threshold on a smaller recognized subset. An exact-path-spelling sensitivity leaves Flat unchanged and recognizes 48 Card tasks, with median 161; the removed Card call names a different `__init__.py` and explicitly fails to replace its text. Both measures recognize mutation-like commands, including failed calls and incidental mentions, rather than certifying successful source edits.

Protocol diagnostic. Two literal thinking/tool markers occur inside submitted actions on 12/99 Flat tasks and 19/99 Card tasks; their paired observations contain syntax-error or cancellation messages. On one fixed completed task in each of Flat, Card, and Control, 13/15 sampled logged responses contain independent reasoning content, while the stored assistant histories omit those fields. The inspected client also omits them when constructing messages, although this source check is not a capture of transmitted requests. These findings document execution limitations; they do not

establish why the markers appeared, whether reasoning loss caused them, or their effect on repair performance.

Other archived interventions. The extended audit also covers a graph-channel variant, an added-rule variant, and an added-file-list variant. All 99 prediction/verifier chains in each of these three conditions pass its checks, with 12, 13, and 9 recorded first-attempt repairs, respectively; eleven later solver reruns in the rule condition are not substituted. Their actual patch-memory exposure is 99/99, 95/99, and 96/99, with no phase refusals; the latter two meet their registered requirements for rule and file-list presence. Eleven SymPy tasks in the file-list condition receive paths with Matplotlib, Sphinx, or Astropy prefixes, illustrating that delivered candidates need not be useful repository locations. The rule retriever and file-list design used offline metrics on these same 99 targets during method selection. Together with runtime failures and the retained Flat provenance limitation, this prevents treating the archived counts as a clean confirmatory comparison of memory mechanisms. In a retrospective family of four comparisons against Flat, Holm-adjusted exact McNemar p values are 1.000 (Card), 0.094 (Graph), 0.453 (Rules), and 0.047 (file list). The file-list difference is -9.09 pp with a paired bootstrap interval $[-16.16, -3.03]$ percentage points; this interval is not adjusted for multiple comparisons. The observed difference does not isolate which aspect of the intervention or execution caused it. Figure 2 shows the paired estimates for both fixed task universes.

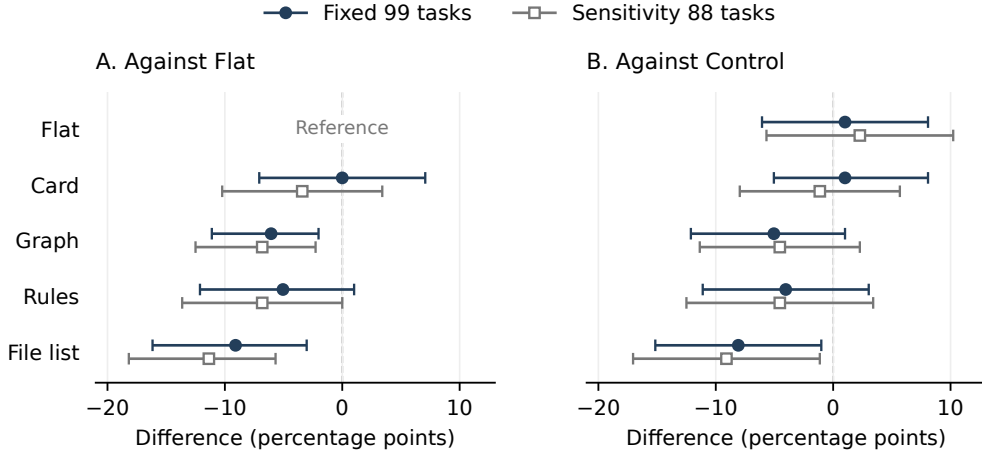


Figure 2: Archived Related-Lite first-prediction differences relative to Flat (left) and the contemporaneous Control (right). Bars are unadjusted 95% paired percentile-bootstrap intervals (200,000 draws; seed 42), not inversions of the exact McNemar tests. These retrospective comparisons include operational failures and retain the historical receipt limitation; they do not isolate causal memory effects.

Control comparison. The contemporaneous Control now has 99 checked first-prediction receipts and resolves 17/99. Its exact corresponding trajectories show no enumerated memory markers in initial prompts and no explicit memory calls; this is a direct-exposure check, not a general contamination certificate. Flat and Card each exceed Control by one observed repair ($+1.01$ pp), with respective paired bootstrap intervals $[-6.06, 8.08]$ and $[-5.05, 8.08]$ percentage points and exact $p = 1$ for both. Neither comparison shows a stable gain, and neither establishes equivalence. In the retrospective five-comparison Holm family against Control, adjusted p values are 1.000 (Flat), 1.000 (Card), 0.906 (Graph), 1.000 (Rules), and 0.287 (file list). The file-list percentile interval excludes zero but does not override its exact test ($p = 0.057$) or that correction. The complete historical sensitivity list leaves Control at 16/88, Flat at 18/88, and Card at 15/88. Control has 48 hard `exit_cost` labels, versus 32 for Flat and 41 for Card; these labels can denote the call cap. We retain runtime failures and empty submissions, never substitute later solver runs, and do not reuse the historical 14/99 Control.

The same original test-summary recognizer covers 61/99 Control tasks, with conditional median first-match index 134. A deterministic source check finds that unittest can report a positive test count

for an artificial `_FailedTest` when the named test cannot be loaded. A uniform 297-trajectory sensitivity excludes observations with enumerated loader-placeholder markers: recognized coverage is then 61/99 Control, 78/99 Flat, and 66/99 Card, with conditional medians 146, 116.5, and 97.5. This preserves the original statistics and all repair outcomes; the revised summaries still do not certify test-body execution or overall speedup.

Separate adaptive Card-v2 revision. A subsequently completed revision changes the environment-check and conditional-repair instructions using the same historical tasks. All 99 original prediction chains pass the frozen receipt audit; Card-v2 resolves 16/99, versus 18/99 for both Flat and Card-v1. Its differences are -2.02 pp, with paired 95% intervals $[-8.08, 3.03]$ against Flat and $[-9.09, 5.05]$ against Card-v1. Exact p values are 0.727 and 0.774; both become 1.000 after Holm correction within these two comparisons. The same 88-task sensitivity gives 16/88 for Card-v2, 18/88 for Flat, and 15/88 for Card-v1. Exact repository cards reach 97/99 targets with no phase refusals; the two unserved attempts remain included. The 39 hard `exit_cost` labels fail the registered maximum of 32, without establishing 39 dollar-budget overruns. The completion analysis was attached after the arm started and partial outcomes were visible. This adaptive result does not establish a repair gain or a causal effect of the revised instructions; its statistics are supplied separately from the original six-condition bundle.

5 VERIFIED 350/150 EXPERIMENT

Split and eligibility. This campaign is separate from Related-Lite. We pin the official 500-task Verified dataset and deterministically allocate 350 training and 150 test tasks using repository-stratified largest-remainder allocation and a fixed hashed ordering (seed 42). The two ID sets are disjoint. The split is within repositories and is not chronological; it does not establish that test tasks are absent from the base model’s training history or earlier project experiments. One issue-text near-duplicate pair is documented before formal solving; excluding its test member defines a fixed 149-task sensitivity analysis. The primary universe remains all 150 tasks.

Memory construction. Only training issues and their official gold patches are used to derive rules. Each extracted record carries source identity and evidence references, with mechanical coverage and evidence-line checks. The complete bank contains 459 rules from 350 sources and 61 mechanism tags. Passing these checks does not prove that every generalized rule is semantically valid; semantic overgeneralization remains a limitation. A deterministic training-only sample of 30 rules received a source-reading AI audit: 20 were judged supported and 10 partly supported, with core edits grounded but errors in explanation or scope. Examples include widening an exact Boolean predicate to truthiness and reversing the explanation of which join comparison includes a filtered condition. These advisory judgments are neither blinded human annotations nor a correctness estimate for the full bank; the frozen bank and evaluation packets remain unchanged. Embeddings are computed locally with a pinned 384-dimensional MiniLM model, without additional embedding API calls. The solver weights remain fixed.

Three conditions. Control receives no reference memory. Flat and Graph share the same rule corpus, dense search, lexical search, rank fusion, diversity reranking, top-three limit, and text budget. Graph additionally enables the Personalized PageRank channel over the fixed rule/source/mechanism structure. All memory packets are constructed from public test descriptions before formal solver calls, then frozen; there are no test-time memory writes. Each packet is inserted once into the initial task input. The saved local configurations apply no history processors. A reconstruction of all 14 attempted formal slots and three separate training attempts preserves the initial task message at every recorded action prefix; all nine formal and both training memory attempts retain their full packets. This checks input retention, not repair efficacy or every wire request. This narrows the estimand to the effect of enabling the graph channel in this implementation, not the effect of every component in the broader ContextGraph system.

The frozen packet contrast confirms a real intervention: the ordered rule IDs differ on 94/150 tasks, and the top-three sets differ on 87/150. All 150 tasks remain in the primary comparison, including those receiving identical packets. Restricting the reported result to tasks whose retrieval changes would answer a different question and is not the primary analysis.

Execution controls. All three conditions use SWE-agent 1.1.0 and the exact DeepSeek V4 Pro endpoint, with no alternate-model fallback. The frozen per-attempt limits are 500 completed calls and a \$30 conservative budget-meter threshold, with equal generation settings and preserved reasoning history. A \$35 per-attempt reservation accounts for the post-response check; the OpenLux route has no provider-enforced price filter. The campaign’s \$300 admission allowance includes \$20 for monitoring and may leave the campaign incomplete. Task blocks contain all three conditions; their six possible orders occur 25 times each across the 150 tasks. The first block uses one worker to measure resources, after which up to four may run if the host and campaign budget permit. No task is admitted without a reservation for its complete three-condition block.

Solver containers receive only public task fields, the frozen memory packet, and a sealed base-commit checkout. The Git seal preserves the base tree and its ancestors while removing future or unreachable commit objects. The solver’s Docker network has no external egress. Credentials remain on the host, and evaluation labels are consumed only by the host-side verifier. Each submission is immediately evaluated with the pinned official SWE-bench 4.1.0 harness and its task-specific original image. Verifier containers have the same two-CPU/eight-GiB cap across all conditions; grading logic is unchanged. Runtime, configuration, tool bundle, image, and source hashes are checked before and after execution. Ten-minute reviews inspect trajectories and execution progress without editing the frozen treatment.

Analysis plan. The primary contrast is Graph minus Flat, and the secondary contrast is Flat minus Control. We use two-sided exact McNemar tests with Holm correction over these two registered comparisons, and paired-task bootstrap confidence intervals with 200,000 replicates and seed 42. We report outcomes against the full 150-task universe, plus the predeclared 149-task sensitivity set. Raw provider usage and a separate conservative budget meter are retained; unreconciled estimates do not establish cost efficiency. Other repository or task slices are exploratory. Budget or infrastructure interruption preserves all artifacts and leaves the campaign incomplete; it does not shrink the primary denominator.

Current evidence. The 350-source bank and 450 memory packets are complete. Training-only real-agent runs exercise memory delivery, the provider protocol, and the official verifier before formal deployment. Their task is excluded from the smoke memory bank and their outcomes are not test-set observations. All 150 images and 450 task/condition inputs are frozen; the original provider passed all three training-only gates before an account-credit error paused launch. A subsequently authorized OpenLux route uses the same DeepSeek V4 Pro snapshot alias as the archived experiments, preserving the split, packets, images, and task order in a separate runtime version. Its tool/reasoning probes and all three training protocol/delivery checks passed. On the excluded training task, Control’s original prediction was officially resolved; Flat’s empty patch and Graph’s reproduction-only patch were unresolved. The original scheduler closed at 12:52 UTC with two complete blocks: all three conditions resolve Django11740 and none resolves Xarray6992. A seventh attempt, Django16256 Flat, autosubmits an empty patch after consecutive command timeouts and is officially unresolved; its outer status classification stops new-block admission. The original registered analysis remains incomplete and has no paired effect estimate. A separately recorded amendment then resumes only previously unstarted slots, accepting ordinary agent-stop outcomes when the original protocol, delivery, prediction and verifier audits pass. It retains all seven original attempts and terminal states, the 150-task denominator, and the original runtime, inputs, limits and budget policy. This post-start interpretation change is disclosed separately; any eventual combined analysis is not an untouched confirmatory experiment. The amendment closed at 14:05 UTC after four complete blocks when a Sphinx Control response supplied a null optional array argument that crashed the agent’s tool parser. Its 47 provider calls pass the protocol audit, but the generic error stop remains inadmissible and the official receipt records an empty patch. Fourteen original slots were attempted, thirteen pass the full audit, and 436 remain unstarted; no failed attempt is replaced. A separately frozen parser correction passes all three training protocol/delivery gates: Control and Flat resolve the excluded training task, while Graph submits an empty patch. All three runs record zero optional-null omissions; the recorded-call regression supplies coverage of the original defect. The formal test runtime and its original outcomes remain unchanged. OpenLux sometimes omits cost and reports non-additive token counts, so raw counters and conservative budget estimates are stored separately without a billing claim. Ten-minute reviews continued until verification ended; incomplete semantic reviews remain recorded.

6 DISCUSSION AND LIMITATIONS

The completed audit rules out a particular ambiguity: the card intervention’s lack of measured gain cannot be explained by widespread absence of any memory response, because actual delivery is observed. However, the reconstructed original responses are transient, and delivery alone does not establish sustained availability. The audit also leaves open incorrect relevance judgments, stale environments, provider variation and insufficient repair skill. Two frozen Pytest test patches already contain their candidate fixes; Matplotlib’s target test also passes before candidate application (Appendix D).

A training-only case illustrates conditional transfer. Flat and Graph both retrieve two rules from Requests1724, whose Python 2 Unicode-method repair adds `builtin_str(method)`. The excluded training task Requests2317 instead reports a Python 3 bytes-method failure at that line; its official training patch replaces the call with `to_native_string(method)`. The retrieved rules already state the Python 2 condition. In the new Graph training run, the agent observes the correct conversion output but ultimately submits an empty patch at its budget limit. Flat, which receives both source rules as well, is officially resolved on the same training task. This documents version dependence and a gap between observing a relevant operation and submitting a repair; it does not establish that memory caused the failure or that an applicability filter improves outcomes. A narrow, case-motivated scan of all 350 training patches recovers this pair only; this is neither an independently sampled case nor an estimate of prevalence.

The new Verified design reduces several confounds by sharing the rule corpus, packet size limits, model settings, and task order policy while varying one retrieval channel. It still has important limits: a single model endpoint, one stochastic attempt per condition, a within-repository random split, official training-patch supervision, and potential historical task exposure. The new protocol also changes multiple execution details relative to historical experiments. Consequently, a future difference from an old score would not isolate the effect of memory.

A stronger causal account would require an independent environment-by-memory intervention, repeated attempts or additional model backbones, and process measures that are validated against real test execution. Such work should use training/development evidence for method selection and freeze evaluation eligibility before inspecting outcomes. The aspiration to improve memory should not make success the inclusion criterion for evidence.

REPRODUCIBILITY STATEMENT

Frozen task lists, source and runtime hashes, per-task official receipts, and the evidence map support the reported claims. Five local anonymous supplements reproduce 18 Related-Lite paired comparisons, 594 exposure checks, 198 retention records with 1,426 processor-boundary checks, all 594 original prediction/report joins, and 287 frozen status comparisons. Delivery uses actual responses and cards; retention uses source-bound structural histories. The verification package contains 302 nonempty predictions with original official reports and 292 empty-prediction exemptions, retaining the historical log exception. These packages do not rerun tests or the complete runtime/Docker audits. The patch-overlap supplement replays Appendix A from hashed features. Credentials and host-only evaluation labels are excluded from agent inputs. The complete anonymous experimental release and human author review remain to be completed.

AI USE STATEMENT

Generative AI tools were used to assist literature discovery and interpretation, formulate and refine hypotheses, design experiments, implement and test code, extract training-memory rules, audit trajectories, analyze results, and draft this manuscript. DeepSeek V4 Pro is also the experimental solver and the source-rule extraction model in the new Verified campaign. Automated progress reviews are advisory; official harness receipts determine repair outcomes. This draft has not yet completed human author review of all AI-assisted claims and artifacts. The final disclosure must reflect the review actually performed before submission.

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A FULL-UNIVERSE PATCH-OVERLAP AUDIT

We retrospectively compare the reference patches actually recorded in all seven conditions, retaining every original task irrespective of its outcome. The source extraction joins 693 original trajectory identities and 641 bound tool-response hashes to the existing delivery audits, yielding 3,020 patch blocks on 585 task-condition records. Separately, we compare all 291 patches in the observed source-pool snapshot with all 99 pinned Related-Lite target patches (28,809 pairs). The snapshot was collected on September 7, 2026; it does not alone attest historical availability.

The primary text screen takes the maximum per-block Jaccard between sets of nonblank added-line contents, stripping surrounding whitespace and excluding diff file headers. It ignores line order and multiplicity and may overstate similarity for short/common edits. We also parse hunk bodies and compare signed changes and file/sign/line features retaining indentation. All threshold counts below use same-repository sources; allowing all repositories gives the same counts.

Table 2: Targets with observed patch overlap, with all counts against the fixed 99-task universe. The first row measures source-pool availability; other rows measure recorded delivery. Missing memory is not certified clean knowledge. No repair outcomes select these records.

Scope	Received patches	$J \geq 0.5$	$J \geq 0.7$	$J = 1$
Full source-pool snapshot	—	18	12	10
Control	0	0	0	0
Flat	99	17	11	8
Card	99	17	11	8
Graph	99	17	11	8
Rules	95	14	8	5
File list	96	17	11	8
Card-v2	97	17	11	8

Seven targets outside the historical exclusion list reach $J \geq 0.5$ in every memory condition. One, Django-34525, shares 38 of 44 normalized added lines in the union with source Django-16820 ($J = 0.864$); its file/sign comparison is 39/46. For Sympy-26642, the one shared added line is the complete Python-file fix; the other is a contributor entry, as the exact-hunk follow-up shows (Appendix B). The historical 88-task sensitivity remains unchanged and is not a complete source-overlap certificate.

No returned source ID belongs to the 99 target IDs, and no block contains its target’s entire patch verbatim. This does not preclude identical edits with different context or hunk locations. The current source patch, after trimming outer whitespace, appears literally in 2,939 blocks; 81 lack that full-patch match, including 64 with interrupted hunks. Text-screen and recognized-hunk added features agree on all observed blocks, but incomplete delivery remains explicit. One pinned target patch, Sympy-27868, fails both our hunk parser and Git’s read-only syntax check at line 319; its strict-hunk comparisons are unknown, while its original bytes and task membership are retained. The file/sign threshold of 0.5 yields 16 observed targets in each memory condition except Rules (13).

The hashed-feature supplement reproduces every pair and task record using only the Python standard library. It does not re-extract private histories or repeat source capture. Lexical overlap does not identify unauthorized access, actual use or a causal contribution to a repair; legitimately related tasks can share fixes. Cards, rule prose, test hints, pretraining and unrecorded channels remain outside this patch-block certificate. All original verifier outcomes are preserved, with no new task exclusion or effect estimate.

B EXACT PATCH-HUNK CORRESPONDENCE

A retrospective follow-up compares complete file hunks across the same 28,809 source–target pairs and 693 original records, after inspecting the seven lexical flags. It retains path headers, hunk coordinates, context, order, indentation and no-newline markers; Git object hashes and mode metadata are outside this equality claim. The Python-file comparison includes every changed .py file and its complete hunks, including comments and docstrings. Empty Python-file sets and malformed patches remain unknown. No repair outcomes select pairs.

Nineteen targets have the same nonempty Python-file sets and exact hunks as a source. Ten match on every changed file; the other nine differ only by additional target documentation, changelog or contributor-metadata files. Every source file’s hunks match its target counterpart in these 19 pairs. To certify recorded delivery here, the complete current source patch must occur literally in a bound response; matching IDs or partial blocks do not suffice.

Table 3: Exact hunk matches, retaining all 99 targets. Counts measure source-pool availability or confirmed complete delivery, not successful repairs. Missing or partial delivery remains explicit.

Scope	All changed files	All Python files
Current source snapshot	10	19
Control	0	0
Flat	8	16
Card	8	16
Graph	8	16
Rules	5	13
File list	8	16
Card-v2	8	16

All seven new lexical flags have exact Python-hunk matches in the pool. Six have confirmed complete matching delivery in every memory condition; Django-34525’s full source-patch delivery remains unconfirmed. An additional target, Matplotlib-23172, exactly matches source Matplotlib-23299 in `lib/matplotlib/__init__.py`, including the backend-preservation fix in `rc_context`. Its extra release note reduces added-line Jaccard to $3/8 = 0.375$, below the earlier threshold; all six memory conditions received the complete matching source. Thus whole-patch lexical thresholds can miss exact implementation fixes when auxiliary files differ.

SWE-ContextBench explicitly includes related issues sharing a resolving pull request (Zhu et al., 2026). These exact matches can arise from the benchmark’s intended construction; they do not establish unauthorized access or causal use by an agent. A separate, incomplete literal replay is described in Appendix C. The original scores and historical 88-task sensitivity remain unchanged. The follow-up’s hashed-file replay reproduces every pair and record using isolated standard-library Python; Git’s weaker stable patch-ID check independently agrees on all 19 Python-file pairs. Human semantic review remains pending.

C LITERAL MEMORY REPLAY AND VERIFIER READINESS

A separate diagnostic freezes all 99 targets in lexicographic order. For each target, it takes the first source block from the first recorded Flat `query_memory` response. It submits the exact current source patch only if the complete patch occurs in that block and the repositories match; otherwise it submits an empty patch. Later sources, later queries, target gold patches, overlap scores and repair outcomes cannot select a substitute. This yields 95 nonempty predictions and four empty predictions. Queries and source order inherit earlier solver trajectories, so this is a trajectory-conditioned replay, not an independent agent benchmark or a claim of zero historical inference cost. It makes no new model calls.

The replay uses the archived verifier code, offline containers and immediate per-task verification. Its new case files use every pinned parquet field, with missing values represented as JSON null. Comparison with the archived cases found 31 field differences across 25 tasks: 25 nullable hint differences, five version differences, and one gold-patch difference. For Sympy-27868, four Unicode line/paragraph separators in the archived patch are LF characters in the parquet; Git accepts the former diff and rejects the latter. The transformation’s upstream cause is unknown. The malformed parquet patch remains unchanged, and these new case files do not replace archived evaluation inputs.

The first run stopped after seven attempts, retaining 92 unstarted tasks and every original receipt. Astropy-4973’s official false verdict explicitly reports a verifier setup failure: its offline image lacks build dependencies, so the submitted patch was not validly tested. The runner checked receipt integrity but failed to reject this readiness error, admitting five further tasks before manual inspection stopped it. The original flag and a separate retrospective readiness classification are both preserved. Among the retained per-task reports, Django-1891’s literal source Django-13315 receives an official resolved verdict. No overall replay repair rate or effect estimate is calculated from this incomplete run. A corrected runner separately checks verifier readiness and stops on explicit infrastructure errors. Original agent scores, task universes and the stopped replay remain unchanged.

An isolated dependency-only image allowed Astropy to import but still yielded zero baseline tests; the corrected runner stopped on that explicit setup error. A further image precompiled the existing extensions without changing the tracked source diff. Initial compilation took 147.8 seconds, exceeding the verifier’s 120-second inner pytest limit. With the same case, source patch and verifier code, this diagnostic then completed: the target test changed from failed to passed and all 55 preservation tests passed. Both earlier failures are retained, and this known-failure diagnostic is not pooled into a replay score. The frozen harness did not retain raw pytest output for the zero-result attempt, limiting attribution of its exact exit cause. After the further diagnostics below, a fresh, separately frozen full-99 replay starts with five declared prepared-image substitutions and mandatory raw-output checks. The seven-attempt run remains closed; the new run is ongoing and supplies no complete repair rate here.

D FURTHER READINESS DIAGNOSTICS AND TEST-PATCH SCOPE

Five additional single-task diagnostics retain the same literal predictions, canonical cases and frozen verifier, with no new model calls. They neither replace the stopped seven-attempt replay nor supply a pooled repair rate. The original Matplotlib image’s final report says only that the candidate patch failed to apply. Newly captured subprocess output also shows an earlier reinstall failure and two pytest collection errors: offline rebuilding cannot download FreeType, leaving the installed testing module without a helper added by the test patch. A separate image caches the exact FreeType and Qhull archives named by the original build metadata and limits compilation to two jobs. Reinstallation then succeeds and all 195 requested statuses are collected. The candidate still fails to apply, and the target test already passes before its application. There is no after-patch test result for this candidate.

Preapplied implementation changes. A static audit compares both patch fields of all 99 canonical cases with the pinned parquet. An explicit path rule identifies Python files outside test directories and filenames in exactly two test patches, Pytest-10341 and Pytest-9358. The paths are `src/_pytest/pathlib.py` and `src/_pytest/tmpdir.py`; their gold patch fields contain only changelog entries. Every file hunk of each already selected literal candidate exactly matches a hunk sequence in its `test_patch`, including coordinates and context. The verifier applies these test patches before the candidates. After separate offline dependency preparation, all three requested target tests are reported passed before candidate application; raw output confirms the corresponding test methods. Both candidates then fail Git application, consistent with their fixes having already been applied. These failures do not establish that the remembered changes were irrelevant. Pytest-10341 also has two other baseline failures, retained in the raw output. The verifier’s existing test-name matching and pretend-version override remain unchanged. The upstream cause of the field split is unknown; the path screen is not a semantic certificate for all other files. No field, task membership, exclusion or original score is changed.

A verified source-patch transfer. A separate scikit-learn image installs Cython 0.29.37 and pre-compiles its existing checkout with two jobs, retaining an empty tracked source diff. Build and import take 203.1 seconds. The unchanged source-25500 patch is then evaluated on target-25763. Its target test changes from failed to passed and all 45 preservation tests pass, with matching raw pytest output and official report. This selected known-failure diagnostic establishes one executable transfer; it does not estimate prevalence, isolate the environment repair’s cause or identify a causal advantage over another agent condition.

Admission and evidence. Report-only readiness had missed the first Matplotlib diagnostic’s intermediate failures because its final report lacked an infrastructure-error label. The original flag is retained. A new runner checks captured output hashes and rejects explicit install, pytest collection/invocation, timeout and signal failures before admitting another task. It distinguishes ordinary failing tests and patch-application errors. Retrospective checking rejects the first Matplotlib attempt and accepts the other four captured sequences; this does not certify stages never executed after application failure. The wrapper captures byte writes returned by the frozen run/create shim, preserving stdin and exit status; inherited OS writes from other Docker verbs are outside scope. Twenty focused software checks pass. All five services are closed, and the local copies match 449 remote file hashes in total. Each package’s source-image and input-preservation checks pass, while all original failures and the full 99-task denominator remain intact.

E HISTORICAL TARGET-TEST TRANSITIONS

The already-passing target tests in the new readiness diagnostics motivate a posthoc check of all 594 original records from the six-condition experiment. This check replays the previously frozen status audit exactly, including its receipt/report joins, before classifying each recorded target status before and after a prediction. It runs neither a solver nor a verifier and preserves the original outcomes and denominators. The 292 empty submissions and 15 setup failures remain distinct from 287 complete comparisons.

None of the 287 complete comparisons records every target test passed before the candidate. All 87 original official resolved labels therefore contain at least one target not recorded passed beforehand. In 71 of these records, every target is recorded `FAILED` before and `PASSED` after. The other 16 include `ERROR` and, in one case, `MISSING` beforehand; neither status is silently treated as an observed test failure.

Condition	Complete comparisons	Official resolved	All targets initially failed
Control	45	17	14
Flat	63	18	14
Card-v1	54	18	16
Graph	56	12	9
Rules	37	13	10
File list	32	9	8

The last column counts resolved records whose every target initially has the exact status `FAILED`. All 16 resolved records outside this group are Django records, and none uses the status audit's baseline fallback. The 1,622 target-test occurrences repeat tasks across conditions and are not independent observations.

This census uses the archived parsed status maps, not complete original pytest stdout. It does not certify the new prepared images, dismiss the two Pytest field anomalies, change the known preservation-test regressions, or establish a causal memory advantage. The new literal replay, its readiness diagnostics, adaptive Card-v2 and the separate Verified experiment remain outside this historical census. Their baseline witnesses do not replace historical maps, and historical maps do not overwrite their new failures. The reproduction plan binds the existing status bundle and the new classifier; a second execution reproduces all classifications and counts.